

2024-11-21-EconNetSimplex

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December 31, 2024

Slide 1

Content

Economic Interpretation of Network Simplex

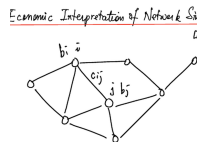
[C] p. 297.

[scale=0.8] [circle, draw, inner sep=2pt, label=above: $b_i \tilde{u}$] (A) at (0,1) ; [circle, draw, inner sep=2pt] (B) at (-1.5,0) ; [circle, draw, inner sep=2pt] (C) at (0,-1) ; [circle, draw, inner sep=2pt] (D) at (1.5,0) ; [circle, draw, inner sep=2pt, label=right: $j \quad b_j$] (E) at (1,-0.5) ; [circle, draw, inner sep=2pt] (F) at (1.5,-1.5) ; [circle, draw, inner sep=2pt] (G) at (3,0.5) ; (A) - (B) - (C) - (A) - (D) - (E) - (F) - (C) - (E) - (D); (A) - (E); (E) - (G); at (0.5,0) c_{ij} ;

Description

This slide presents an economic interpretation of the network simplex method in the context of linear programming. The title "Economic Interpretation of Network Simplex" is underlined. A reference "[C] p. 297" is given, likely referring to a specific page in a cited textbook or other source. The main content is a hand-drawn network graph. The nodes represent points in a network, and the edges represent connections between them. Each node is labeled, with some nodes having labels like $b_i \tilde{u}$ and $j \quad b_j$, suggesting these labels represent supply or demand at each node. The edge between nodes labeled $b_i \tilde{u}$ and $j \quad b_j$ is labeled c_{ij} , which likely represents the cost or some other economic parameter associated with that connection. The overall graph structure suggests a transportation or flow network problem, where the network simplex method would be used to find an optimal flow through the network. The context of linear programming is crucial because the network simplex method is a specialized algorithm for solving linear programs with a network structure. The hand-drawn nature of the graph suggests this is a lecture slide or a note taken during a lecture.

Figures



(a) Hand-drawn network graph with nodes and edges labeled.

Slide 2

Content

Economic Interpretation of Network Simplex

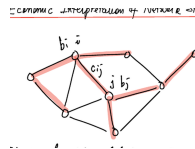
[scale=0.8] [circle, draw, inner sep=2pt, label=above: $b_i \tilde{u}$] (A) at (0,1) ; [circle, draw, inner sep=2pt] (B) at (-1.5,0) ; [circle, draw, inner sep=2pt] (C) at (0,-1) ; [circle, draw, inner sep=2pt] (D) at (1.5,0) ; [circle, draw, inner sep=2pt, label=right: $j \ b_j$] (E) at (1,-0.5) ; [circle, draw, inner sep=2pt] (F) at (1.5,-1.5) ; [circle, draw, inner sep=2pt] (G) at (3,0.5) ; [red!30, line width=2pt] (A) – (B); [red!30, line width=2pt] (A) – (E); [red!30, line width=2pt] (E) – (D); [red!30, line width=2pt] (E) – (C); (A) – (C); (B) – (C); (C) – (D); (C) – (F); (D) – (E); (E) – (G); (D) – (F); at (0.5,0.2) c_{ij} ;

Start a feasible solution on a spanning tree

Description

This slide discusses the economic interpretation of the network simplex method within the context of linear programming. The title, "Economic Interpretation of Network Simplex," is written at the top. A network graph is drawn, with nodes representing locations and edges representing connections between them. Some nodes are labeled with variables like $b_i \tilde{u}$ and $j \ b_j$, likely representing supply or demand at those locations. The edge between nodes $b_i \tilde{u}$ and $j \ b_j$ is labeled c_{ij} , which probably represents the cost associated with that connection. A subset of the edges are highlighted in a light red color, forming a spanning tree. The sentence "Start a feasible solution on a spanning tree" indicates that the network simplex algorithm begins by finding a feasible solution that corresponds to a spanning tree of the network. This is a key characteristic of the network simplex method, which leverages the special structure of network problems to improve efficiency. The context of linear programming is crucial because the network simplex method is a specialized algorithm for solving linear programs with a network structure, such as transportation or assignment problems. The hand-drawn nature of the graph suggests this is a lecture slide or notes from a lecture.

Figures



(a) Hand-drawn network graph with nodes and edges labeled. Some edges are highlighted in a light red color.

Slide 3

Content

Economic Interpretation of Network Simplex

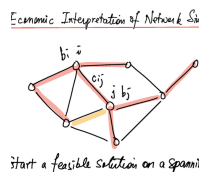
[scale=0.8] [circle, draw, inner sep=2pt, label=above: $b_i \tilde{u}$] (A) at (0,1) ; [circle, draw, inner sep=2pt] (B) at (-1.5,0) ; [circle, draw, inner sep=2pt] (C) at (0,-1) ; [circle, draw, inner sep=2pt] (D) at (1.5,0) ; [circle, draw, inner sep=2pt, label=right: $j \ b_j$] (E) at (1,-0.5) ; [circle, draw, inner sep=2pt] (F) at (1.5,-1.5) ; [circle, draw, inner sep=2pt] (G) at (3,0.5) ; [red!30, line width=2pt] (A) – (B); [red!30, line width=2pt] (A) – (E); [red!30, line width=2pt] (E) – (D); [red!30, line width=2pt] (E) – (C); (A) – (C); (B) – (C); (C) – (D); (C) – (F); (D) – (E); (E) – (G); (D) – (F); at (0.5,0.2) c_{ij} ;

Start a feasible solution on a spanning tree

Description

This slide illustrates the economic interpretation of the network simplex method in linear programming. The title "Economic Interpretation of Network Simplex" is written at the top. A network graph is presented, where nodes represent locations and edges represent connections between them. Some nodes are labeled with variables like $b_i \tilde{u}$ and $j \ b_j$, likely representing supply or demand at those locations. The edge between nodes $b_i \tilde{u}$ and $j \ b_j$ is labeled c_{ij} , which probably represents the cost associated with that connection. A subset of the edges are highlighted in a light red color, forming a spanning tree. The sentence "Start a feasible solution on a spanning tree" indicates that the network simplex algorithm begins by finding a feasible solution that corresponds to a spanning tree of the network. This is a key characteristic of the network simplex method, which leverages the special structure of network problems to improve efficiency. The context of linear programming is crucial because the network simplex method is a specialized algorithm for solving linear programs with a network structure, such as transportation or assignment problems. The hand-drawn nature of the graph suggests this is a lecture slide or notes from a lecture. The question mark "(?)" suggests an incomplete thought or a point to be discussed further.

Figures



(a) Hand-drawn network graph with nodes and edges labeled. Some edges are highlighted in a light red color.

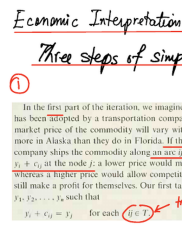
Slide 4

Content

Description

This slide presents the first step in the economic interpretation of the network simplex method for solving linear programming problems, specifically in the context of transportation networks. The title "Economic Interpretation of Network Simplex" is underlined. The subtitle "Three steps of simplex" is also underlined. The main content describes an iterative process where a feasible tree solution is assumed to be adopted by a transportation company. The shipping costs cause the market price of the commodity to vary by location. The core concept is to determine a set of "fair prices" $(y_1, y_2, \dots, y_n)$ such that the equation $y_i + c_{ij} = y_j$ holds for each arc (ij) in the feasible tree (T) . This equation ensures that the price at node j (y_j) is equal to the price at node i (y_i) plus the shipping cost from i to j (c_{ij}). This prevents wasteful shipping (lower price at j) and underselling by competitors (higher price at j). The circled "1" indicates this is the first of three steps in the simplex method. The context of linear programming is crucial because the network simplex method is a specialized algorithm for solving linear programs with a network structure.

Figures



(a) Handwritten notes describing the economic interpretation of the network simplex method, focusing on the first step of the simplex algorithm. The notes explain how to determine a set of "fair prices" for a transportation company based on shipping costs and market prices at different locations. A circled "1" indicates this is the first step. The equation $y_i + c_{ij} = y_j$ is central to this step, where y_i represents the unit price at node i , c_{ij} is the shipping cost from node i to node j , and y_j is the unit price at node j . The set T likely represents the set of arcs in the current feasible tree solution.

Slide 5

Content

Economic Interpretation of Network Simplex

Three steps of simplex

2

In the second step of each iteration, we put ourselves in the role of a competitor: would it pay to buy the commodity at some node i , ship it along an arc ij and sell at the node j ? In our example, the answer is affirmative for several different arcs ij . For instance, we could buy at the node 2 for $y_2 = 23$ and ship along the arc 25 for $c_{25} = 37$: the total expense of $y_2 + c_{25} = 60$ compares favorably with the selling price of $y_5 = 62$. In general, this second step consists of choosing an arc ij such that

$$y_i + c_{ij} < y_j$$

non-tree arc

$$i \xrightarrow{-z_{ij} < 0} j$$

$$y_j - y_i + z_{ij} = c_{ij} \quad \text{i.e.} \quad y_j - y_i > c_{ij}$$

Description

This slide details the second step of the network simplex algorithm within the context of a linear programming course focused on network flows. The title "Economic Interpretation of Network Simplex" and subtitle "Three steps of simplex" are written at the top. The main content describes an iterative process where the algorithm simulates a competitor's decision-making process. The competitor evaluates whether buying a commodity at node i , shipping it along arc (i, j) , and selling it at node j is profitable. This is determined by comparing the total cost $(y_i + c_{ij})$ with the selling price (y_j) at node j . The inequality $y_i + c_{ij} < y_j$ indicates a profitable opportunity. The equation $y_j - y_i + z_{ij} = c_{ij}$ is also presented, likely representing a constraint or condition within the simplex method. The circled "2" indicates this is the second step of a three-step process. The context of linear programming is crucial because the network simplex method is a specialized algorithm for solving linear programs with a network structure, such as transportation or assignment problems. The handwritten style suggests this is a lecture slide or notes from a lecture.

Figures

Economic Interpretation
Three steps of simplex
 ②

In the second step of each iteration, we put would it pay to buy the commodity at some i the node i ? In our example, the answer is affirmative, we could buy at the node 2 for $y_2 =$

(a) Handwritten notes describing the second step of the network simplex method. The notes explain how to determine if it is profitable to buy a commodity at one node, ship it to another, and sell it there, based on the current prices and shipping costs. A circled "2" indicates this is the second step. The inequality $y_i + c_{ij} < y_j$ is central to this step, where y_i represents the price at node i , c_{ij} is the shipping cost from node i to node j , and y_j is the price at node j . The equation $y_j - y_i + z_{ij} = c_{ij}$ is also present, where z_{ij} is a variable related to the flow along arc (i, j) . The circled "i" and "j" likely represent nodes in the network.

Slide 6

Content

Economic Interpretation of Network Simplex

Three steps of simplex

3

In the third step of each iteration, we imagine that the first company has found out about its competitor's plan and hastens to use the information to its own advantage. It will now ship t units through the entering arc and keep ignoring other out-of-tree arcs; the shipments along the arcs of T will have to be adjusted so as to maintain feasibility of the resulting schedule. The adjustments, easy to work out, are shown in

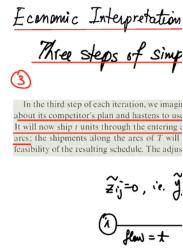
$$z_{ij} = 0, \quad \text{i.e.} \quad \tilde{y}_j - \tilde{y}_i = c_{ij}$$

$[scale = 0.8][circle, draw, innersep = 2pt](A)at(0,0); [circle, draw, innersep = 2pt](B)at(2,0); [->](A) -- (B)node[midway, above]flow = t;$

Description

This slide presents the third and final step of the network simplex method's economic interpretation within the context of a linear programming course focused on network flows. The title "Economic Interpretation of Network Simplex" and subtitle "Three steps of simplex" are written at the top. The main content describes a scenario where a company learns about its competitor's shipping plan and adjusts its own plan accordingly. The company ships t units through a new arc (entering arc), while adjusting shipments along existing arcs to maintain feasibility. The equation $\tilde{z}_{ij} = 0$, or equivalently $\tilde{y}_j - \tilde{y}_i = c_{ij}$, represents the condition for maintaining feasibility after the adjustment, where $\tilde{y}_i$ and $\tilde{y}_j$ are the adjusted prices at nodes i and j , and c_{ij} is the cost of shipping from i to j . A simple diagram illustrates the flow of t units from node i to node j . The circled "3" indicates this is the third step of a three-step process. The context of linear programming is crucial because the network simplex method is a specialized algorithm for solving linear programs with a network structure, such as transportation or assignment problems. The handwritten style suggests this is a lecture slide or notes from a lecture.

Figures



(a) Handwritten notes describing the third step of the network simplex method. The notes explain how a company would adjust its shipping plan based on its competitor's actions. A circled "3" indicates this is the third step. The equation $\tilde{z}_{ij} = 0$, or equivalently $\tilde{y}_j - \tilde{y}_i = c_{ij}$, is central to this step, where $\tilde{y}_i$ and $\tilde{y}_j$ represent adjusted prices at nodes i and j , and c_{ij} is the shipping cost from i to j . A simple diagram shows the flow of t units from node i to node j .